- In high latitudes, salinity increases with depth. In the middle latitudes, it increases up to 35 metres and then it decreases. At the equator, surface salinity is lower.

Statement 3: INCORRECT—Salinity, generally, increases with depth and there is a distinct zone called the halocline (compare this with thermocline), where salinity increases sharply.

12. Statement 1: CORRECT—When the sun, the moon and the earth are in a straight line, the height of the tide will be higher. These are called spring tides and they occur twice a month, one on full moon period and another during new moon period.

Statement 2: INCORRECT—The most common tidal pattern, featuring two high tides and two low tides each day is said to be Semi-diurnal tide.

Statement 3: CORRECT—Once in a month, when the moon's orbit is closest to the earth (perigee), unusually high and low tides occur. During this time the tidal range is greater than normal.

Two weeks later, when the moon is farthest from earth (apogee), the moon's gravitational force is limited and the tidal ranges are less than their average heights.

13. Characteristics of Tides
    - The tidal bulges on wide continental shelves have greater height.
    - In the open ocean tidal currents are relatively weak.
    - When tidal bulges hit the mid-oceanic islands they become low.
    - The shape of bays and estuaries along a coastline can also magnify the intensity of tides.
    - Funnel-shaped bays greatly change tidal magnitudes. Example: *Bay of Fundy — Highest tidal range.*
    - The large continents on the planet, however, block the westward passage of the tidal bulges as the Earth rotates.
    - Tidal patterns differ greatly from ocean to ocean and from location to location.
14. Statement 1: CORRECT—Fringing reefs are reefs that grow directly from a shore. They are located very close to land, and often form a shallow lagoon between the beach and the main body of the reef.

Statement 2: INCORRECT—Barrier reefs are extensive linear reef